

Physics-Informed Bandgap Engineering in Quantum Dots: A Theoretical and Computational Framework for Core, Shell, and Material Simulation

Fundamental Physics of Quantum Confinement and Material Composition

The electronic and optical properties of semiconductor quantum dots (QDs) are fundamentally governed by spatial quantum confinement, which manifests when the physical dimensions of a nanocrystal become comparable to or smaller than the exciton Bohr radius (a_B) of the parent bulk material¹. In bulk semiconductors, electronic states form continuous energy bands separated by a bulk bandgap ($E_{g, \text{bulk}}$)¹. When spatial confinement is imposed in three dimensions, the continuous translational symmetry of the crystal lattice is broken, causing the conduction and valence bands to collapse into discrete, atomic-like energy levels⁴. This spatial quantization blueshifts the effective optical bandgap (E_g) relative to $E_{g, \text{bulk}}$ as the particle radius (R) decreases¹.

The baseline electronic structure of a quantum dot is primarily dictated by its constituent elemental composition². Elemental composition determines intrinsic material constants, including $E_{g, \text{bulk}}$, the conduction band electron effective mass (m_e^*), the valence band hole effective mass (m_h^*), spin-orbit splitting energy (Δ_{so}), and the static relative permittivity (ϵ_r)¹. The exciton Bohr radius serves as the fundamental spatial scale defining the onset of quantum confinement dynamics and is expressed as:

$$a_B = \epsilon_r \left(\frac{m_0}{\mu} \right) a_0$$

where $a_0 \approx 0.0529 \text{ nm}$ is the Bohr radius of the hydrogen atom, m_0 is the free electron rest mass, and μ is the reduced exciton effective mass defined by $\mu^{-1} = (m_e^*)^{-1} + (m_h^*)^{-1}$.

Varying the chemical composition shifts both $E_{g, \text{bulk}}$ and the spatial scale a_B , fundamentally altering the sensitivity of E_g to radius modifications². In II-VI binary semiconductors (such as CdSe, CdS, and ZnS), increasing anion electronegativity or decreasing cation atomic radius lowers the valence band maximum (VBM) and raises the conduction band minimum (CBM), expanding $E_{g, \text{bulk}}$ ². Concurrently, wider bulk bandgap semiconductors exhibit heavier effective masses and lower dielectric constants, which depresses a_B and necessitates smaller physical radii to reach strong quantum confinement². Conversely, IV-VI lead chalcogenides (such as PbS and PbSe) possess narrow direct bulk bandgaps located at the high-symmetry L -point of the Brillouin zone⁸. These materials feature symmetric, light effective masses ($m_e^* \approx m_h^* \approx 0.09m_0$) alongside high static dielectric constants ($\epsilon_r > 15$)⁸. Consequently, a_B in lead chalcogenides expands beyond **18 nm**, placing even relatively large nanocrystals ($R \approx 5 - 8 \text{ nm}$) deep within the strong confinement regime and enabling broad optical tunability across the near-to-shortwave infrared spectrum⁸. III-V semiconductors (such as InP) exhibit strong covalent bonding with a high degree of wavefunction delocalization, yielding light electron effective masses ($m_e^* \approx 0.07m_0$) and moderate exciton Bohr radii ($a_B \approx 9.6 \text{ nm}$)¹¹. Metal halide perovskites (such as CsPbBr_3 and CsPbI_3) display defect-tolerant band structures where the VBM is formed by anti-bonding hybrid states of lead $6p$ and halide p orbitals, while the CBM is formed by lead $6p$ states³. Compositional halide substitution ($\text{Br}^- \rightarrow \text{I}^-$) modulates the lattice constant and orbital overlap, systematically tuning $E_{g, \text{bulk}}$ from **2.25 eV** down to **1.73 eV** while preserving moderate effective masses and exciton radii ($a_B \approx 3.5 - 5.6 \text{ nm}$)³.

The interplay between dielectric screening (ϵ_r) and carrier effective masses (μ) establishes a critical transition threshold between strong and weak confinement regimes across different material systems¹. In high-dielectric, low-mass materials, the kinetic confinement energy scales faster than the Coulombic electron-hole attraction, rendering single-particle quantization energy the dominant term in determining the bandgap¹.

Size-Dependent Confinement Dynamics and

Analytical/Numerical Methods

Analytical Formulation: The Brus Equation and Particle-in-a-Box

The basic analytical model describing size-dependent optical bandgap shifts in spherical quantum dots is the Brus equation, derived within the framework of the Effective Mass Approximation (EMA) under infinite potential barrier boundary conditions¹. The bandgap $E_g(R)$ as a function of dot radius R is expressed as:

$$E_g(R) = E_{g, \text{bulk}} + \frac{\hbar^2 \pi^2}{2R^2} \left(\frac{1}{m_e^*} + \frac{1}{m_h^*} \right) - \frac{1.8e^2}{4\pi\epsilon_0\epsilon_r R} - 0.248E_R^*$$

where $\hbar$ is the reduced Planck constant, e is the elementary charge, ϵ_0 is the vacuum permittivity, ϵ_r is the high-frequency or static dielectric constant of the semiconductor, and E_R^* is the effective Rydberg energy¹.

The physical contributions to the Brus equation are divided into distinct physical interactions¹:

- **Bulk Energy Term** ($E_{g, \text{bulk}}$): Represents the baseline electronic bandgap of the unconfined bulk crystal¹.
- **Quantum Confinement Kinetic Energy Term** ($\propto R^{-2}$): Derived from the ground-state ($1s$) kinetic energy of an electron and a hole confined inside a three-dimensional spherical infinite potential well¹. The R^{-2} scaling forces a rapid expansion of the bandgap as radius decreases¹.
- **Coulomb Interaction Term** ($\propto R^{-1}$): Accounts for the electrostatic Coulomb attraction between the co-confined electron and hole wavefunctions, attenuated by dielectric screening within the nanocrystal¹. This term shifts the exciton transition energy downward¹.
- **Spatial Correlation Energy Term (Effective Rydberg Correction)**: A size-independent or weakly size-dependent polarization correction representing higher-order spatial correlation effects¹.

Higher-Order Physics and Breakdown of the Brus Equation

While the Brus equation offers clear intuitive utility, it exhibits substantial quantitative discrepancies when evaluated against experimental optical absorption spectra for small

nanocrystals ($R < 2 \text{ nm}$)⁶. This failure stems from three primary physical oversimplifications inherent to the standard EMA model⁴:

1. **Infinite Potential Barrier Assumption:** Real quantum dots possess finite potential barriers dictated by the electron affinity and ionization potential of surrounding matrix materials or capping organic ligands⁴. Wavefunctions penetrate into the barrier region, softening the kinetic confinement energy and reducing the R^{-2} scaling slope⁴.
2. **Band Non-Parabolicity:** The parabolic dispersion relation $E(k) = \frac{\hbar^2 k^2}{2m^*}$ holds strictly near the Brillouin zone center (Γ -point)⁶. Strong quantum confinement shifts carrier wavevectors high into the band structure, where non-parabolicity increases the carrier effective mass with energy, dampening the bandgap expansion⁶.
3. **Position- and Size-Dependent Effective Mass:** Effective masses are not strictly constant bulk parameters; they undergo spatial variation near the particle boundary and experience renormalization in ultrasmall clusters due to lattice relaxation and altered orbital hybridization⁶.

To quantify these regimes for computational modeling, quantum confinement dynamics are categorized by physical scale¹:

Confinement Regime	Radial Criteria	Prevailing Physics	Dominant Computational Model
Weak Confinement	$R \gg$	Center-of-mass exciton quantization; Coulomb energy dominates kinetic confinement ¹ .	Perturbative EMA with Excitonic Mass $M = m_e^* + \quad$ ¹ .
Strong Confinement	$R \ll$	Independent carrier quantization; kinetic energy dominates Coulomb attraction ¹ .	Finite-Barrier EMA / Multi-Band $k \cdot p$ Theory ⁶ .
Extreme Confinement (EC)	$R \ll$ (Ultrasmall / Clusters)	Discrete atomic orbital states; thermal expansion and lattice vibrations diverge from bulk ¹ .	Tight-Binding (TB) / Empirical Pseudopotential (EPM) ³ .

In the extreme confinement regime, physical parameters such as the temperature coefficient

of the bandgap ($\frac{dE_g}{dT}$) deviate sharply from bulk values¹. Deriving $\frac{dE_g}{dT}$ from the temperature derivative of the modified Brus equation highlights these size-dependent thermal expansion and phonon-coupling contributions¹:

$$\frac{dE_g}{dT} = \frac{dE_{g, \text{bulk}}}{dT} + \left[-\frac{\hbar^2 \pi^2}{3R^2} \left(\frac{1}{m_e^*} + \frac{1}{m_h^*} \right) + \frac{1.8e^2}{12\pi\epsilon_0\epsilon_r R} \right] \alpha_V + \frac{\hbar^2 \pi^2}{2R^2} \frac{\partial}{\partial T} \left(\frac{1}{m_e^*} + \frac{1}{m_h^*} \right)$$

where α_V is the volumetric thermal expansion coefficient of the crystal lattice¹.

Atomistic and High-Fidelity Computational Methodologies

To eliminate the overestimation of bandgaps inherent to simple EMA models, advanced computational techniques are required for simulator algorithms³:

- **Finite-Depth Square-Well EMA:** Incorporates finite potential barrier heights (V_0) at the quantum dot interface, solving transcendental boundary equations⁴. This model captures wavefunction evanescent decay outside the physical dot radius, providing excellent agreement with experimental data down to $R \approx 1 \text{ nm}$ ⁴.
- **Multi-Band $k \cdot p$ Perturbation Theory:** Solves coupled differential Hamiltonians (such as 6×6 or 8×8 Kane models) that implicitly account for non-parabolicity, valence band degeneracy (heavy-hole, light-hole, split-off mixing), and spin-orbit coupling⁶.
- **Empirical Pseudopotential Method (EPM):** Atomistic approach replacing envelope functions with linear combinations of plane waves acted upon by semi-empirical atomic pseudopotentials fit to experimental bulk band structures³. EPM naturally captures surface geometries, lattice contractions, non-parabolic dispersion across the entire Brillouin zone, and indirect-to-direct band transitions in ultrasmall nanocrystals¹².
- **Tight-Binding (TB) Method:** Expresses wavefunctions as linear combinations of atomic orbitals with parameterized hopping integrals between nearest and next-nearest neighbors⁶. TB models successfully calculate anisotropic effective masses, atomistic surface reconstruction, and ligand passivation effects in non-spherical or distorted nanocrystals⁶.

Core/Shell Heterostructure Engineering and Interface Dynamics

Band Alignment Classifications

Synthesizing a secondary semiconductor shell over a core nanocrystal introduces heterojunction band offsets that allow spatial engineering of electron and hole wavefunctions⁷.

Depending on the relative energy levels of the isolated bulk core and shell materials

(conduction band offsets ΔE_c and valence band offsets ΔE_v), core/shell QDs are divided into three electronic alignments²¹:

Alignment Type	Relative Band Offsets	Wavefunction Spatial Localization	Optical & Electronic Consequences
Type-I	$E_{g, \text{shell}} >$; positive ΔE_c and ΔE_v ¹¹ .	Both electron and hole wavefunctions localized inside the core ¹⁷ .	High photoluminescence quantum yield; surface defect passivation; core-dominated emission tuning ⁵ .
Type-II	Staggered bands; core CBM/VBM both higher or lower than shell ¹⁷ .	Complete spatial separation: one carrier in core, the other in shell ²¹ .	Redshifted transition energies below bulk bandgaps; extended exciton radiative lifetimes ²¹ .
Quasi-Type-II	One large offset ($\Delta E_v \gg$), one minimal offset ($\Delta E_c \approx$) ⁷ .	One carrier localized in core; other carrier delocalized across core/shell ⁷ .	Continuous redshifting with shell thickness; suppressed non-radiative Auger recombination ⁷ .

In Type-I heterostructures (such as CdSe/ZnS and InP/ZnS), the shell acts as a potential barrier, passivating surface dangling bonds, suppressing non-radiative surface recombination, and enhancing photoluminescence quantum yield while maintaining size-tunable core emission⁵. In Type-II structures (such as CdTe/CdSe and ZnSe/CdS), spatial separation reduces the electron-hole overlap integral, redshifting the optical transition energy below the bulk bandgap of either constituent material and extending radiative lifetimes²¹. In Quasi-Type-II systems (such as CdSe/CdS), increasing shell thickness systematically expands the effective confinement volume for the delocalized carrier, continuously tuning the optical bandgap and suppressing Auger rates⁷.

Quantum Mechanical Interface Boundary Conditions

Solving single-particle wavefunctions across heterostructure interfaces with distinct material

domains requires matching conditions that satisfy quantum mechanical probability current conservation¹⁷. Because the effective mass of charge carriers differs between core (m_{core}^*) and shell (m_{shell}^*) materials, standard derivative continuity breaks down¹⁷. Instead, envelope functions must satisfy the BenDaniel-Duke boundary conditions at the interface boundary $r = R_c$ ¹⁷:

$$\psi_{\text{core}}(R_c) = \psi_{\text{shell}}(R_c)$$

$$\frac{1}{m_{\text{core}}^*} \frac{d\psi_{\text{core}}(r)}{dr} \Big|_{r=R_c^-} = \frac{1}{m_{\text{shell}}^*} \frac{d\psi_{\text{shell}}(r)}{dr} \Big|_{r=R_c^+}$$

Applying BenDaniel-Duke boundary conditions ensures that the normal component of probability current density remains continuous across steep material interfaces, preventing unphysical accumulation of charge probability at step potential barriers¹⁷.

Lattice Mismatch, Strain Dynamics, and Deformation Potentials

Epitaxial growth of shell layers on core nanocrystals with mismatched crystal lattice constants ($a_{\text{core}} \neq a_{\text{shell}}$) introduces coherent mechanical strain². The lattice mismatch parameter is defined as:

$$\varepsilon_{\text{mismatch}} = \frac{a_{\text{shell}} - a_{\text{core}}}{a_{\text{core}}}$$

For small shell thicknesses, the shell stretches or compresses to match the core lattice parameter, generating hydrostatic and shear strain profiles throughout the heterostructure⁶.

Hydrostatic strain induces a local volume change ($\Delta V/V$), which shifts the core and shell band edges via deformation potential theory²:

$$\Delta E_{c,\text{strain}} = a_c \left(\frac{\Delta V}{V} \right)$$

$$\Delta E_{v,\text{strain}} = a_v \left(\frac{\Delta V}{V} \right)$$

where a_c and a_v are the conduction and valence band hydrostatic deformation potentials, respectively².

When a core material with a larger lattice constant is encapsulated by a shell material with a smaller lattice constant (such as a CdSe core with $a = 6.05 \text{ \AA}$ coated by a ZnS shell with $a = 5.41 \text{ \AA}$), the core experiences compressive hydrostatic strain². Compressive strain

expands the bulk bandgap of the core ($E_{g, \text{core}}$), shifting both conduction and valence confinement levels outward and causing an additional blueshift in the effective bandgap². As shell thickness increases beyond a critical threshold (typically 3–5 monolayers), the strain energy exceeds the threshold for dislocation formation, resulting in lattice relaxation, interfacial defects, and optical quenching¹².

Numerical Framework for Quantum Dot Simulator Implementation

To build a high-fidelity quantum dot bandgap simulator engine, the governing differential equations must be solved numerically over a 1D spatial domain using spherical symmetry¹⁷.

Single-Particle Radial Schrödinger Equation

Assuming spherical symmetry, 3D wavefunctions are factorized into radial envelope functions

$R_{nl}(r)$ and spherical harmonics $Y_{lm}(\theta, \phi)$ ²⁶. Defining the reduced radial wavefunction $u_{nl}(r) = r \cdot R_{nl}(r)$, the 1D single-particle Schrödinger equation with a

position-dependent effective mass $m^*(r)$ and radial potential profile $V(r)$ is written as¹⁷:

$$-\frac{\hbar^2}{2} \frac{d}{dr} \left(\frac{1}{m^*(r)} \frac{du_{nl}(r)}{dr} \right) + \left[V(r) + \frac{\hbar^2 l(l+1)}{2m^*(r)r^2} \right] u_{nl}(r) = E_{nl} u_{nl}(r)$$

The position-dependent profile functions for a core/shell/matrix system with core radius R_c and outer shell radius R_s are defined piecewise¹⁷:

$$V_e(r) = \begin{cases} 0 & \text{if } 0 \leq r \leq R_c \text{ (Core)} \\ \Delta E_c & \text{if } R_c < r \leq R_s \text{ (Shell)} \\ V_{0,e} & \text{if } r > R_s \text{ (Matrix/Ligands)} \end{cases}$$

$$m_e^*(r) = \begin{cases} m_{e,\text{core}}^* & \text{if } 0 \leq r \leq R_c \\ m_{e,\text{shell}}^* & \text{if } R_c < r \leq R_s \\ m_{e,\text{matrix}}^* & \text{if } r > R_s \end{cases}$$

Analogous profile functions $V_h(r)$ and $m_h^*(r)$ are constructed for holes using valence band offsets ΔE_v and potential barriers $V_{0,h}$.

Perturbative Coulomb Interaction Integration

Once ground-state radial wavefunctions ($l = 0, n = 1$) for the electron ($u_e(r)$) and hole ($u_h(r)$) are computed along with their single-particle energy eigenvalues (E_e and E_h), the excitonic bandgap $E_{\text{gap, exciton}}$ is determined by subtracting the electrostatic Coulomb attraction integral¹:

$$E_{\text{gap, exciton}} = E_{\text{g, bulk, core}} + E_e + E_h - E_{\text{Coulomb}} + \Delta E_{\text{strain}}$$

The Coulomb correction integral E_{Coulomb} accounts for spatially varying dielectric permittivity $\epsilon_r(r)$ across heterostructure interfaces and is evaluated numerically via¹:

$$E_{\text{Coulomb}} = e^2 \int_0^\infty dr_e \int_0^\infty dr_h \frac{|u_e(r_e)|^2 |u_h(r_h)|^2}{4\pi\epsilon_0\epsilon_r(\max(r_e, r_h)) \max(r_e, r_h)}$$

Self-Consistent Poisson-Schrödinger Algorithmic Sequence

In heavily doped quantum dots, high excitation regimes (biexcitons/multiexcitons), or structures with extreme dielectric contrast, single-particle approximations become insufficient¹⁰.

Simulator accuracy requires a self-consistent Poisson-Schrödinger iterative solver sequence to compute electrostatic potential bending and carrier charging²⁹. The computational loop proceeds through the following sequential execution steps²⁹:

1. **Grid and Domain Initialization:** Input spatial grid discretization ($dr \approx 0.01 \text{ nm}$), material domain boundaries (R_c, R_s), band offsets ($\Delta E_c, \Delta E_v$), effective masses, and dielectric constants¹⁷. Set initial electrostatic potential $V_{\text{electrostatic}}^{(0)}(r) = 0$.
2. **Schrödinger Eigen-Solver Execution:** Discretize the Hamiltonian matrix using a finite difference scheme under BenDaniel-Duke boundary conditions¹⁷. Compute single-particle eigenvalues E_e, E_h and eigenvectors $u_e(r), u_h(r)$ ¹⁷.
3. **Charge Density Calculation:** Calculate spatial carrier charge distributions $\rho(r)$ from carrier occupation numbers N_e and N_h ²⁹:

$$\rho(r) = e \left[N_h \left| \frac{u_h(r)}{r} \right|^2 - N_e \left| \frac{u_e(r)}{r} \right|^2 \right]$$

4. **Poisson Differential Equation Solver:** Solve the 1D spherical electrostatic differential equation for $V_{\text{electrostatic}}(r)$ ²⁹:

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \epsilon_0 \epsilon_r(r) \frac{dV_{\text{electrostatic}}(r)}{dr} \right) = -\rho(r)$$

5. **Potential Profile Update:** Update the confinement potential profile using a linear mixing parameter $\alpha \approx 0.1 - 0.3$ to guarantee numerical convergence stability²⁹:

$$V^{(k+1)}(r) = V_{\text{confinement}}(r) - e \cdot \left[(1 - \alpha) V_{\text{electrostatic}}^{(k)}(r) + \alpha V_{\text{electrostatic}}^{(k+1)}(r) \right]$$

6. **Convergence Checking:** Compare energy eigenvalues between successive iterations⁶. If $|E^{(k+1)} - E^{(k)}| > 10^{-5} \text{ eV}$, repeat execution from Step 2⁶. Upon convergence, output finalized excitonic bandgaps and spatial wavefunctions²⁹.

Material Parameter Database for Simulation Engines

To feed accurate material constants into numerical quantum dot simulation engines, physical parameters compiled across major semiconductor families are tabulated below¹:

Material System	Crystal Structure	Lattice Constant a (Å)	Bulk Bandgap E_g , bulk (eV)	Electron Mass m_e^*/m_0	Hole Mass m_h^*/m_0	Static Permittivity ϵ_r	Exciton Bohr Radius a_B (nm)	VBM Energy Relative to Vacuum (eV)
Cadmium Selenide (CdSe)	Wurtzite / Zincblende	6.05	1.74	0.13	0.45	9.60	5.60	−6.40 [cite: 7, 16, 28]
Cadmium Sulfide (CdS)	Wurtzite / Zincblende	5.82	2.42	0.19	0.80	5.70	2.80	−6.80 [cite: 1, 7, 12, 33]
Zinc Sulfide (ZnS)	Zincblende / Wurtzite	5.41	3.61	0.28	0.53	8.30	1.50	−7.50 [cite: 5, 11, 26]
Lead Sulfide (PbS)	Rocksalt	5.94	0.41	0.09	0.09	17.20	18.00	−5.20 [cite: 8, 9, 10]

Indium Phosphide (InP)	Zincblende	5.87	1.35	0.07	0.60	12.50	9.60	−5.80 [cite: 11, 12]
Cesium Lead Bromide (CsPbBr_3)	Perovskite (Cubic)	5.87	2.25	0.15	0.14	7.00	3.50	−5.85 [cite: 2, 8, 11, 13, 33]
Cesium Lead Iodide (CsPbI_3)	Perovskite (Cubic)	6.28	1.73	0.12	0.11	6.32	5.60	−5.40 [cite: 3, 13, 14]

Strategic Guidelines for Quantum Dot Simulator Implementation

To implement an accurate simulation architecture, software models must incorporate multi-regime physical approximations and numerical matching routines:

- Confinement-Dependent Modeling Selection:** The computational engine should dynamically select mathematical solvers based on R/a_B . For $R/a_B > 2$ (weak confinement), simple analytical Effective Mass Approximations with excitonic mass corrections are sufficient¹. For $0.5 \leq R/a_B \leq 2$ (strong confinement), finite-depth square-well Schrödinger solvers incorporating BenDaniel-Duke boundary conditions are necessary⁴. For $R/a_B < 0.5$ (extreme confinement), atomistic empirical pseudopotentials or multi-band non-parabolic $k \cdot p$ solvers must be deployed to capture band dispersion folding³.

- **Boundary Condition Enforcement:** Numerical finite-difference schemes across core/shell interfaces must explicitly implement BenDaniel-Duke matching conditions ($\frac{1}{m_1^*} \frac{d\psi_1}{dr} = \frac{1}{m_2^*} \frac{d\psi_2}{dr}$)¹⁷. Standard derivative matching introduces significant energy level errors, particularly for heterostructures with large mass mismatches (such as InP/ZnS)¹¹.
- **Coherent Strain Field Integration:** Core/shell modules must calculate isotropic radial and tangential strain profiles resulting from lattice constant mismatch ($a_{\text{shell}} \neq a_{\text{core}}$)². Hydrostatic deformation potential shifts ($\Delta E = a_{\text{hydro}} \frac{\Delta V}{V}$) should be added directly into the local band offset potential arrays prior to solving the Schrödinger equation².
- **Self-Consistent Charge Solver Coupling:** For charging phenomena, multi-excitonic states, or highly polarizable shell systems, single-particle energy states must be iteratively coupled with a 1D spherical Poisson solver²⁹. Updating the electrostatic potential profile dynamically until energy convergence reaches $< 10^{-5} \text{ eV}$ guarantees accurate modeling of spatial carrier separation in Type-II and Quasi-Type-II quantum dots⁶.

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